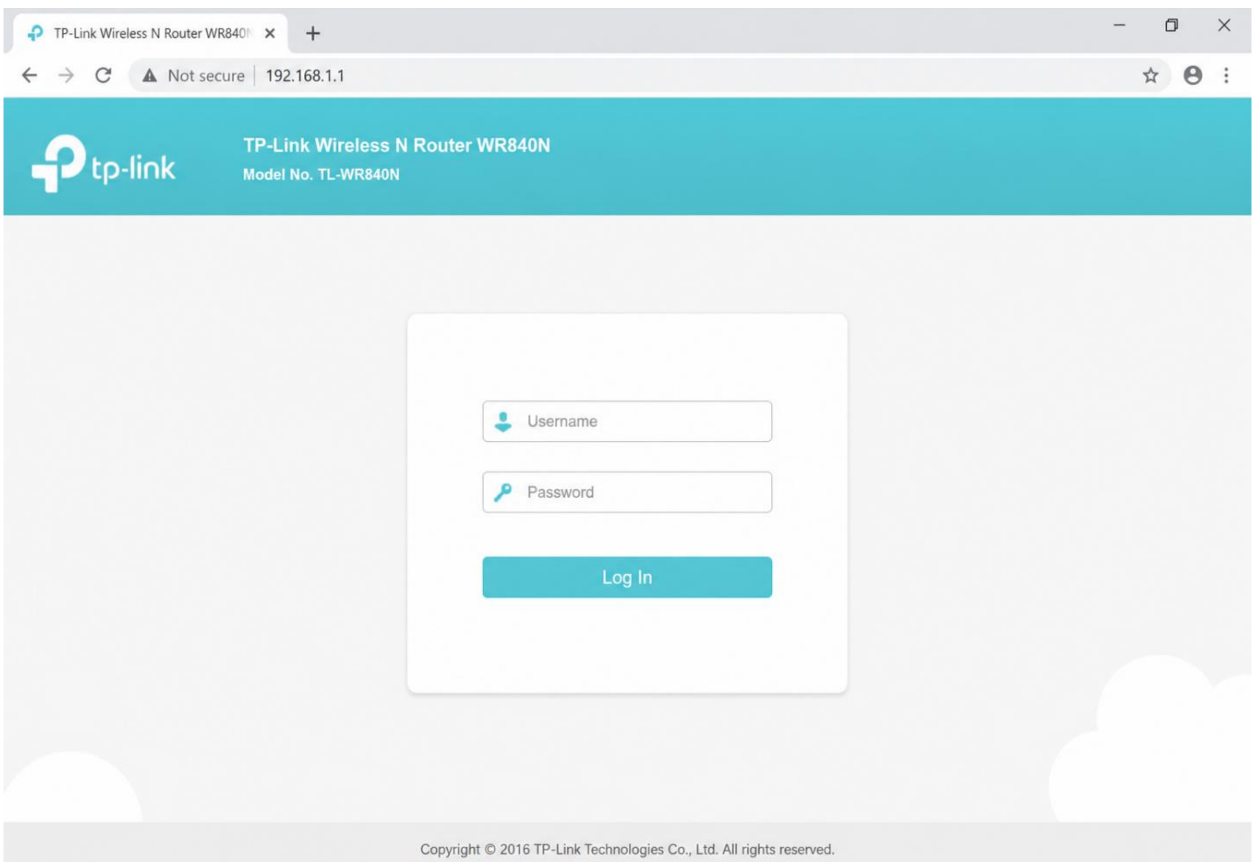
Day 3: Configuring Basic Network Hardware Theory :IP addressing basics and network configuration .Practical :Set up a router with default settings.

Tasks :-

**1.Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.**

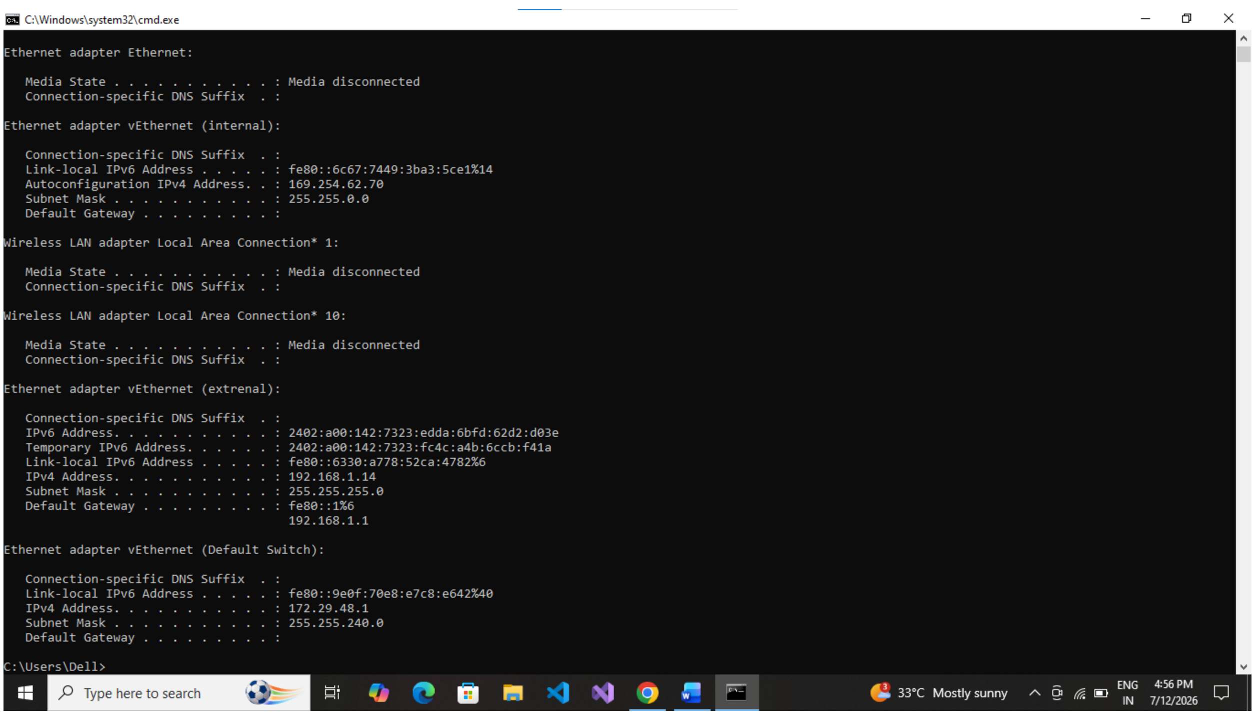
Ans. :-



2.Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Ans. :-

1. Press Windows + R
2. Type cmd and press Enter
3. Type the following command: ipconfig
4. Press Enter.



Current Network Configuration:-

- IP Address (IPv4):- 192.168.1.14
- Subnet Mask:- 255.255.255.0
- Default Gateway:- 192.168.1.1

**3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.**

Ans. :-

Steps to Set a Static IP (Windows 10) :-

1. Press **Windows + R**, type ncpa.cpl, and press **Enter**.
2. Right-click your **Wi-Fi** adapter and select **Properties**.
3. Double-click **Internet Protocol Version 4 (TCP/IPv4)**.
4. Select **Use the following IP address**.
5. Enter :

Setting	**Value**
IP Address	192.168.1.50
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
Preferred DNS Server	8.8.8.8
Alternate DNS Server	8.8.4.4

6. Click **OK** and close all windows.

Test Internet :- cmd in enter;ping google.com

Or ping 8.8.8.8

Revert Back to DHCP (Important) :-

After testing:

1. Open **TCP/IPv4 Properties** again.
2. Select:
    - Obtain an IP address automatically
    - Obtain DNS server address automatically
3. Click **OK**.

4.In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

Ans. :-

DHCP Range :-

Setting	**Value**
Router IP Address	192.168.1.1
DHCP Start IP	(192.168.1.100)
DHCP End IP	(192.168.1.199)

If two devices are assigned the same IP address within the network, an IP address conflict occurs. The router cannot correctly identify which device should receive network traffic, causing one or both devices to lose network or internet connectivity and resulting in an unstable connection.

Effects of an IP Address Conflict :-

- One or both devices may lose internet connectivity.
- Devices may frequently disconnect and reconnect to the network.
- File sharing, printing, and other network services may stop working.
- Windows may display an "IP address conflict detected" warning.
- Network performance becomes slow and unstable because packets may be delivered to the wrong device.
- Applications such as video calls, online gaming, and web browsing may experience interruptions.
- The router may repeatedly update its ARP table as both devices claim the same IP address.